

REPORT
OF
DATA ANALYTICS ESSENTIALS

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.



In partial fulfilment of the requirement of degree of
B.Tech CSE

By

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2026–27

CANDIDATE’S DECLARATION

I, Sreedevi EJ, hereby declare that the report entitled “Data Analytics Essentials” has been prepared by me as part of the academic requirements of the Bachelor of Technology in Computer Science and Engineering at IILM University, Greater Noida. This report presents the learning, concepts, tools and outcomes associated with my completion of the Data Analytics Essentials course offered through Cisco Networking Academy.

I confirm that the material included in this report has been organized for academic submission and that references and supporting documents have been acknowledged wherever applicable. The report has been prepared specifically for this academic requirement and has not been submitted by me to another institution for the award of any other degree or diploma. I understand that I am responsible for the originality and appropriate use of the material included in the report.

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me during the completion of the Data Analytics Essentials course and the preparation of this report. The training gave me an opportunity to study data analytics in a structured manner and to understand how raw data can be prepared, examined and communicated to obtain useful information.

I am thankful to the School of Computer Science and Engineering, IILM University, Greater Noida, for providing an academic environment in which I could develop my technical skills alongside my regular B.Tech CSE studies. I am also grateful to the faculty members and academic mentors whose guidance has helped me understand the importance of applying classroom concepts to practical technical work.

I would also like to acknowledge Cisco Networking Academy for providing the Data Analytics Essentials learning experience. The course introduced me to the overall data analytics process, characteristics of data, data acquisition, data preparation and transformation, basic statistical analysis, and the use of tools such as Excel, SQL and Tableau. These areas helped me understand that data analysis is not limited to producing charts or calculations; it also involves deciding what data is useful, preparing it correctly and communicating the resulting information clearly.

I am grateful to my parents for their constant encouragement and support throughout my studies. Their motivation has helped me remain consistent while completing the course and preparing this report.

Finally, I would like to thank everyone who directly or indirectly contributed to the completion of this work. The knowledge gained from this training will be useful as I continue developing my technical skills in computer science and explore data-oriented applications in future academic and professional work.

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ABSTRACT

Data has become an important resource for organizations because it can support planning, evaluation, problem solving and decision-making when it is collected and analyzed appropriately. The Data Analytics Essentials course offered through Cisco Networking Academy provided a structured introduction to the major stages involved in working with data. The learning covered the data analytics process, characteristics of data, methods of acquiring data, data preparation and transformation, basic statistical analysis, and practical use of analytics tools including Excel, SQL and Tableau.

This report documents the learning undertaken through the course and explains how the different concepts fit together as a data analytics workflow. It discusses the movement of data from its initial collection through cleaning, preparation, transformation, analysis, visualization and communication of findings. Particular attention is given to the role of Excel, SQL and Tableau, since these tools support different stages of practical analytics work. Excel can be useful for organizing data and performing calculations, SQL is suited to querying structured data, and Tableau can be used to communicate patterns and findings visually.

The report also describes the objectives, methodology, scope and learning outcomes associated with the training. Rather than treating analytics as a single software skill, the report presents it as a process that combines data understanding, preparation, analytical reasoning and communication. The course strengthened my understanding of how technical tools can be combined with logical thinking to extract useful information from datasets. The knowledge gained is relevant to my B.Tech CSE studies and provides a foundation for further learning in areas such as business intelligence, data analysis, database systems, visualization and data-driven software applications.

Keywords: Data Analytics, Data Preparation, Excel, SQL, Tableau, Data Visualization, Statistics, Data Transformation, Data Acquisition

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Table 0.1 – List of Abbreviations

Abbreviation	Meaning
DA	Data Analytics
SQL	Structured Query Language
EDA	Exploratory Data Analysis
BI	Business Intelligence
CSV	Comma-Separated Values
KPI	Key Performance Indicator
DB	Database
ETL	Extract, Transform, Load

CHAPTER 1 – INTRODUCTION

1.1 Introduction

Data analytics is the process of examining data in a systematic way to identify useful information, patterns, relationships and trends. In a computing environment, data can be produced by applications, databases, websites, sensors, transactions, surveys and many other sources. The presence of large amounts of data by itself does not guarantee useful decision-making. Data has to be understood, prepared and analyzed before meaningful conclusions can be drawn.

The Data Analytics Essentials course provided a structured introduction to this process. The course focused on how the analytics process creates value from data and introduced important practical areas including data characteristics, data acquisition, transformation, basic statistical techniques and the use of analytics tools. The certificate associated with the course specifically identifies Excel, SQL, Tableau and other tools as part of the hands-on learning areas.

For a computer science student, understanding analytics is useful because data is closely connected with software systems and databases. Applications generate and store information, while analytics techniques can be used to understand that information. The course therefore provides a useful connection between programming, databases, statistics and practical decision-making.

1.2 Background of Data Analytics

The growth of digital systems has increased the amount and variety of data available to organizations. A business may maintain information about customers, products, orders, payments and services. Educational institutions may work with student records, attendance, assessment results and survey responses. Similarly, websites and applications can produce information about user activity. In all of these cases, the usefulness of data depends on its quality and on the ability to interpret it correctly.

Data analytics addresses this need by providing a sequence of activities through which raw information can be converted into useful insights. The sequence generally begins with identifying and acquiring relevant data, followed by preparation and transformation. Once the data is suitable for analysis, statistical and analytical

methods can be applied. Finally, the findings can be communicated through tables, charts, dashboards or written explanations.

An important lesson from the training is that analysis should not begin with a visualization tool alone. A chart may look attractive while still representing incomplete, inconsistent or incorrectly prepared data. Good analytics therefore requires attention to the quality and characteristics of the underlying dataset.

1.3 Importance of Data Analytics

Data analytics is important because it provides a systematic way of moving from observations to evidence-based conclusions. Instead of relying only on assumptions, an analyst can examine available data and identify measurable patterns. This can support activities such as monitoring performance, identifying unusual values, comparing groups, understanding customer behaviour and evaluating outcomes.

Analytics is also important in technical environments because data is often produced continuously by software systems. Knowing how to work with that data allows a computer science professional to contribute beyond software development alone. Skills in data preparation, querying, statistics and visualization can be applied in business intelligence, application development, database administration, product analysis and many other areas.

1.4 Role of Data in Modern Organizations

Data can support different levels of organizational activity. At an operational level, it can help track transactions and routine activities. At a management level, summarized information can support comparisons and performance monitoring. At a strategic level, trends and historical patterns can contribute to planning.

However, the value of data depends on factors such as accuracy, completeness, consistency, relevance and availability. Data that contains duplicate records, missing values or inconsistent formats can lead to misleading results. Therefore, data preparation is not a minor preliminary task; it is an important part of the analytics process.

1.5 Overview of the Data Analytics Essentials Course

The Data Analytics Essentials course covered the major elements required for an introductory understanding of data analytics. The certificate records learning in explaining how analytics creates value, understanding data characteristics and

acquisition methods, transforming data with analytics tools, applying basic statistical and preparation techniques, using Excel, SQL and Tableau through hands-on activities, and evaluating and sharing a project portfolio.

The course therefore combines conceptual understanding with tool-oriented learning. This combination is useful because analytics work requires both an understanding of why a particular activity is performed and the practical ability to perform it with appropriate tools.

CHAPTER 2 – ORGANIZATION / COURSE PROFILE

2.1 About Cisco Networking Academy

Cisco Networking Academy is the learning platform identified on the course completion certificate provided for this report. The certificate confirms completion of the Data Analytics Essentials course and identifies the Cisco Networking Academy as the issuing learning organization.

For the purpose of this report, the relevant focus is the Data Analytics Essentials learning experience rather than a general organizational history. The course provides structured learning around data analytics concepts and tools and is designed to help learners develop an understanding of how data can be transformed into useful information.

2.2 Overview of Data Analytics Essentials

Data Analytics Essentials introduces the learner to the main activities involved in working with data. The learning covers the data analytics process, data characteristics, data acquisition, data transformation, basic statistical and data preparation techniques, and hands-on work with tools such as Excel, SQL and Tableau.

The course is particularly useful at the foundation level because it connects several skills that are often studied separately. Data preparation connects raw data with analysis, SQL connects analytics with databases, Excel supports practical calculations and organization, and Tableau supports visual communication of findings.

Table 2.1 – Main Areas Covered by the Data Analytics Essentials Course

Learning Area	Focus
Analytics process	Understanding how data moves from collection to useful insight
Data characteristics	Understanding formats, availability and properties of data
Data acquisition	Understanding ways in which relevant data can be obtained
Data preparation	Preparing data so that it can be analyzed reliably
Data transformation	Changing data into useful structures or representations
Basic statistics	Using basic statistical ideas to summarize and understand data
Analytics tools	Hands-on exposure to Excel, SQL, Tableau and other tools
Project portfolio	Evaluating and communicating completed analytical work

2.3 Learning Structure

The course can be understood as a progression from fundamental concepts to practical tool usage. First, the learner needs to understand what data is and why analytics creates value. The next step is to understand how data is obtained and what characteristics affect its use. Data then needs to be prepared and transformed before analysis. Finally, analytical results can be communicated through appropriate tools and project work.

This progression is important because each stage depends on the previous stage. Poorly acquired or prepared data can affect the quality of later analysis, while technically correct analysis may still be difficult to understand if it is not communicated clearly.

2.4 Skills Developed During the Course

The training developed an introductory set of technical and analytical skills. These include understanding data workflows, identifying data characteristics, preparing and transforming datasets, applying basic statistical techniques and working with common analytics tools. Another important skill is the ability to evaluate and present analytical work as a portfolio.

From a computer science perspective, these skills complement programming and database knowledge. They encourage a more structured approach to data, in which the learner considers not only how to process information but also whether the information is suitable for the intended analysis.

CHAPTER 3 – PROBLEM STATEMENT AND OBJECTIVES

3.1 Problem Statement

Organizations and software systems generate large amounts of data, but raw data is not automatically useful. Data may contain missing values, inconsistent formats, duplicates, irrelevant fields or values that require transformation. If such data is analyzed without proper preparation, the resulting conclusions may be inaccurate or difficult to trust.

The central problem addressed by data analytics training is therefore how to convert available raw data into information that can be interpreted and communicated effectively. This requires an understanding of the data itself, methods for acquiring it, techniques for preparing and transforming it, suitable analytical methods, and tools for presenting results.

3.2 Motivation

The motivation for learning data analytics comes from the growing role of data in computing and decision-making. As a CSE student, understanding how databases and applications generate information is useful, but being able to analyze that information provides an additional perspective. It makes it possible to identify patterns, compare results and communicate findings instead of only storing or processing records.

The course also provides exposure to tools that are commonly associated with practical analytics workflows. Learning these tools at an introductory level creates a foundation for more advanced study.

3.3 Project / Training Objectives

1. Understand the purpose and overall process of data analytics.
2. Identify important characteristics and formats of data.
3. Understand methods through which data can be acquired for analysis.
4. Learn why data preparation is necessary before analytical work.
5. Understand common data transformation activities.

6. Apply basic statistical ideas for summarizing and examining data.
7. Develop introductory practical skills using Excel for data analysis.
8. Develop introductory skills in using SQL to work with structured data.
9. Understand how Tableau can be used to visualize and communicate analytical findings.
10. Develop the ability to evaluate and communicate analytical work through a project portfolio.

3.4 Learning Objectives

The learning objectives are both technical and analytical. Technically, the objective is to become comfortable with the basic stages of a data workflow and with selected tools. Analytically, the objective is to develop the habit of asking whether data is appropriate, whether it has been prepared correctly and what evidence supports a conclusion.

The course is not treated as a replacement for advanced statistics, machine learning or specialized data engineering. Instead, it establishes a foundation on which those areas can be studied later.

CHAPTER 4 – PROJECT / TRAINING DESCRIPTION

4.1 Introduction

The training can be viewed as a complete introductory workflow for handling data. The process starts with a question or requirement, followed by identifying the required data and obtaining it. The data is then checked and prepared. After preparation, it can be transformed into a suitable form and analyzed using basic statistical methods or other analytical techniques. The results can then be visualized and communicated.

The order matters. For example, if a dataset contains dates in several formats, calculating time-based trends before standardizing those dates can produce incorrect results. Similarly, duplicate records can affect counts and averages. Data analytics therefore combines technical operations with careful reasoning.

4.2 Data Analytics Process

The data analytics process is a sequence rather than a single operation. A typical workflow begins by defining the problem or question. The next step is to identify

relevant data sources and acquire the required data. Once available, the data is inspected for structure and quality. Preparation activities such as handling missing values, correcting inconsistent formats and removing unnecessary records may then be performed.

After preparation, transformation can be used to create fields or structures required for analysis. The prepared dataset can then be summarized or examined using statistical methods. Visualization provides a way to communicate important patterns. Finally, the findings need to be interpreted in relation to the original question.

Table 4.1 – Stages of the Data Analytics Process

Stage	Purpose	Typical Output
Define	Clarify the question or requirement	Analytical question
Acquire	Obtain relevant data	Raw dataset
Prepare	Check and improve data quality	Clean dataset
Transform	Create analysis-ready structures	Transformed dataset
Analyze	Identify patterns and summarize data	Statistics/findings
Visualize	Communicate important patterns	Charts/dashboards
Interpret	Relate findings to the original question	Insights
Communicate	Share results with appropriate context	Report/presentation

4.3 Characteristics of Data

Before analyzing a dataset, it is necessary to understand what the data represents and how it is structured. Data may be numerical, categorical, textual, date-based or a mixture of different types. It may also be organized in tables, files or other structures. Availability and accessibility are additional considerations because useful data cannot be analyzed if it cannot be obtained appropriately.

The characteristics of data also affect the choice of analytical method. Numerical values can be summarized using measures such as averages, while categorical values

may be examined through counts and proportions. Dates can support trend analysis when they are stored consistently. Understanding these differences reduces the risk of applying unsuitable operations.

Table 4.2 – Common Characteristics of Data

Characteristic / Type	Description	Example
Numerical	Values representing quantities	Sales amount, marks
Categorical	Values representing groups or labels	Department, product type
Text	Free-form or descriptive information	Comments, descriptions
Date/Time	Values representing time	Order date, registration date
Structured	Data organized into predictable fields	Relational table
Unstructured	Data without a fixed tabular structure	Text documents

4.4 Data Acquisition

Data acquisition refers to obtaining the information needed for an analysis. The appropriate source depends on the question being investigated. Data may be available in files, databases, application exports, surveys or other sources. The quality of the acquisition process matters because incorrect or incomplete collection can affect every later stage.

An analyst should also consider whether the acquired fields are relevant to the question. Collecting large amounts of unnecessary information can increase preparation effort without improving the analysis. Therefore, data acquisition should be guided by the purpose of the analysis.

4.5 Data Preparation

Data preparation involves checking and improving the dataset before analysis. Typical issues include missing values, duplicate records, inconsistent spelling, incorrect data types, invalid entries and inconsistent date or numerical formats. Preparation does not necessarily mean deleting every unusual value. Instead, each issue needs to be considered in relation to the meaning of the data.

For example, a missing value may represent information that was not collected, while an extreme numerical value may be a legitimate observation rather than an error. Treating every unusual value as incorrect can introduce bias. Good preparation therefore requires both technical checks and an understanding of the dataset.

4.6 Data Transformation

Transformation changes data into a form that is more suitable for analysis. Examples include converting data types, standardizing labels, deriving new fields, grouping values, combining information and restructuring tables. Transformation can make patterns easier to analyze without changing the underlying meaning of the information.

For example, individual transaction dates can be transformed into month or year fields to support trend analysis. Numerical values may be grouped into ranges when a categorical comparison is more useful than individual values. The important point is that transformations should be documented so that the analysis remains understandable and reproducible.

4.7 Exploratory Data Analysis

Exploratory Data Analysis, or EDA, involves examining a dataset to understand its main characteristics and identify patterns or unusual observations. At an introductory level, this can include checking counts, minimum and maximum values, averages, distributions and relationships between variables.

EDA is useful because it helps the analyst form a better understanding of the data before making conclusions. Visualizations can support this process by making differences and trends easier to notice. However, visual patterns should still be interpreted carefully and should not automatically be treated as proof of causation.

4.8 Basic Statistical Analysis

Basic statistics provides a way to summarize numerical data. Measures such as mean, median, minimum, maximum and range can help describe a dataset. Counts and percentages are useful for categorical information. The choice of measure should depend on the type and distribution of the data.

For example, the mean provides an arithmetic average, while the median represents the middle value when observations are ordered. When a dataset contains extreme values, the median may sometimes provide a more representative description

of a typical observation. Even at a basic level, understanding the difference between these measures is important for responsible interpretation.

4.9 Excel for Data Analysis

Excel is useful for introductory analytics because it provides a familiar tabular environment for organizing data, applying formulas, filtering records and creating charts. A learner can inspect a dataset, calculate summary measures and create visual representations without needing to build a complete software application.

In a practical workflow, Excel can be used to identify missing values, sort and filter records, calculate totals or averages, and prepare simple summaries. Its usefulness is particularly strong for small to medium-sized datasets and for situations where users need to inspect information interactively. At the same time, large or complex analytical workflows may require databases or specialized tools.

4.10 SQL for Data Analysis

SQL, or Structured Query Language, provides a way to retrieve and work with data stored in relational databases. For analytics, SQL can be used to select relevant records, filter information, group values and calculate aggregate results. This makes SQL an important bridge between database systems and data analysis.

A simple analytical query might group sales records by product category and calculate a total for each group. More advanced queries can combine related tables and apply conditions. The main advantage is that the analysis can be performed close to the database, reducing the need to manually export every record into another tool.

4.11 Tableau for Data Visualization

Tableau is a visualization and analytics tool that can be used to represent data through charts, dashboards and interactive views. Visualization is important because analytical results are often easier to communicate through a well-designed chart than through a long table of values.

An effective visualization should match the analytical question. A line chart may be appropriate for a trend over time, while a bar chart can support comparison between categories. A dashboard can combine several related views to provide a broader picture. Good visualization also requires attention to labels, units, scale and context so that the audience does not misinterpret the chart.

4.12 Project Portfolio Evaluation

The course also included evaluation and sharing of project portfolio work. A portfolio can demonstrate not only the final chart or result but also the process used to reach it. A strong analytical portfolio should make the problem clear, describe the data, explain the preparation and analysis, and communicate the findings in a way that another person can understand.

For a student, portfolio-based work is useful because it provides evidence of practical learning. It also encourages the learner to explain technical decisions rather than simply showing an output.

CHAPTER 5 – TECHNOLOGIES AND TOOLS USED

5.1 Microsoft Excel

Excel supports data organization, calculations, filtering, sorting and visualization in a spreadsheet environment. During introductory analytics work, it can be used to inspect rows and columns, calculate summary values, identify patterns and create basic charts. It is also useful for manually reviewing a dataset before moving to more specialized tools.

5.2 SQL

SQL is used for interacting with relational data. In analytics, its main role is to retrieve and summarize the required records efficiently. Filtering, grouping and aggregation are particularly useful when an analyst needs to answer questions from structured datasets. SQL also develops an important understanding of how analytical work relates to database structures.

5.3 Tableau

Tableau is used to communicate data visually. Its role in an analytics workflow is especially relevant after data has been prepared and meaningful questions have been identified. The analyst can select appropriate visual forms and build views that allow patterns and comparisons to be communicated more clearly.

5.4 Other Analytics Tools

The course certificate refers to Excel, SQL, Tableau and other tools. Since the supplied course certificate does not provide a complete list of those additional tools,

this report does not assign specific unverified tools to the training. The focus is therefore kept on the tools explicitly identified in the certificate.

Table 5.1 – Technologies and Tools Used

Tool	Primary Role	Typical Use
Excel	Spreadsheet analysis	Cleaning, calculations, filtering, summaries and charts
SQL	Database querying	Selecting, filtering, grouping and aggregating structured data
Tableau	Visualization	Charts, dashboards and visual communication

Table 5.2 – Comparison of Excel, SQL and Tableau

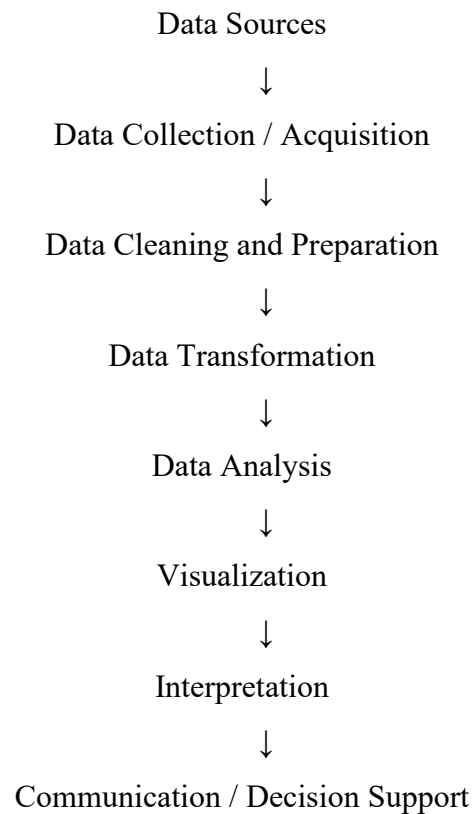
Aspect	Excel	SQL	Tableau
Main purpose	Tabular analysis	Database querying	Visualization
Best suited for	Small/medium analytical tasks	Structured database data	Visual exploration and communication
Strength	Accessible and flexible	Efficient data retrieval	Interactive visual presentation
Typical output	Tables, formulas, charts	Query results	Charts/dashboards
Place in workflow	Preparation and analysis	Acquisition/retrieval and analysis	Visualization and communication

CHAPTER 6 – SYSTEM ARCHITECTURE / DATA ANALYTICS WORKFLOW

6.1 Overview

For this training report, system architecture is represented as a data workflow rather than a software application architecture. The purpose is to show how information moves through different stages before it becomes an analytical insight. Each stage has a specific responsibility and produces an output that supports the next stage.

Figure 6.1 – Overall Data Analytics Workflow



6.2 Data Collection

The first operational stage is the collection or acquisition of relevant data. The source must be appropriate to the analytical question. At this stage, it is useful to record where the data came from, what each field represents and whether there are limitations in coverage or completeness.

6.3 Data Cleaning and Preparation

After collection, the data is checked for quality. This can include identifying missing values, duplicate rows, inconsistent labels, invalid entries and incorrect formats. The purpose is not simply to make the dataset look clean but to reduce errors that could influence the analysis.

6.4 Data Transformation

Transformation prepares the data for the specific analytical question. Derived columns, groups, standardized labels and other representations can be created when required. The transformation stage should preserve the meaning of the source information and be documented clearly.

6.5 Data Analysis

Analysis involves applying appropriate calculations or analytical methods to the prepared data. Basic statistical summaries can identify central values and variation, while grouping and comparison can reveal differences between categories. The analysis should remain connected to the original question rather than generating statistics without a clear purpose.

6.6 Data Visualization

Visualization turns selected analytical results into graphical representations. The objective is not to create as many charts as possible but to choose visual forms that make relevant patterns easy to understand. A good visualization should have meaningful labels and should avoid creating misleading impressions through inappropriate scales or unnecessary decoration.

6.7 Interpretation and Communication

The final stage is to interpret the results and communicate them to an intended audience. Interpretation involves explaining what the results show and, equally importantly, what they do not establish. Communication can take the form of a report, presentation, dashboard or portfolio item depending on the context.

CHAPTER 7 – METHODOLOGY

7.1 Learning Methodology

The training was approached as a combination of conceptual learning and practical tool exposure. Concepts were considered first so that tool operations could be understood in context. The learning then moved toward data preparation, transformation, analysis and visualization. This approach helps avoid treating individual software commands as isolated skills.

7.2 Data Analysis Methodology

A suitable methodology begins with a clearly defined question. Once the question is known, the analyst determines what information is required and identifies an appropriate data source. The acquired data is inspected and prepared before analysis. Appropriate calculations are then performed and results are visualized for interpretation.

This approach can be summarized as:

1. Define the analytical requirement.
2. Identify relevant data.
3. Acquire the data.
4. Inspect its structure and quality.
5. Clean and prepare the data.
6. Transform fields where necessary.
7. Analyze using suitable techniques.
8. Visualize important findings.
9. Interpret results in context.
10. Communicate the outcome clearly.

7.3 Data Preparation Process

Table 7.1 – Data Preparation Activities

Activity	Purpose	Example Check
Structure check	Understand columns and records	Identify field names and data types
Missing-value check	Identify incomplete information	Count blank entries
Duplicate check	Avoid repeated records	Compare key fields
Format check	Maintain consistency	Standardize dates or labels
Validity check	Identify impossible entries	Check value ranges
Relevance check	Remove unnecessary information	Confirm fields support the question

7.4 Statistical Analysis

Basic statistical analysis provides a compact description of data. Depending on the dataset, this may include counts, percentages, averages, medians, minimum and maximum values and ranges. These measures can be used to compare groups or understand the overall distribution.

Statistics should be interpreted in context. A single average does not describe every observation, and a correlation or visible pattern does not automatically demonstrate causation. Even basic analysis therefore requires careful reasoning.

7.5 Visualization and Interpretation

Visualization was treated as a communication stage of the analytical workflow. The choice of chart depends on what needs to be communicated. Comparisons, trends, distributions and relationships can require different visual forms. The analyst should also consider who will view the result and what decision or question the visualization is intended to support.

7.6 Project / Portfolio Approach

A portfolio-oriented approach encourages the learner to present the problem, data, process and result as a connected story. This makes the analytical work easier to evaluate. It also develops communication skills because a portfolio must explain technical work to someone who may not have performed the analysis themselves.

CHAPTER 8 – LEARNING OUTCOMES AND RESULTS

8.1 Technical Skills Acquired

The course developed a foundational understanding of the data analytics workflow. The most important technical learning was the ability to recognize that data analysis involves multiple stages and that each stage affects the reliability of the final result. The training also provided hands-on exposure to Excel, SQL and Tableau as different tools supporting different parts of the workflow.

8.2 Analytical Thinking

One of the important learning outcomes was a more structured way of approaching data-related questions. Instead of immediately calculating values or creating charts, an analyst should first identify the question, understand the available data and check whether it is suitable. This mindset is useful across many technical domains.

8.3 Data Preparation Skills

The training highlighted the importance of preparing data before analysis. Missing values, inconsistent formats, duplicates and unsuitable fields can affect analytical

results. Understanding these issues provides a practical foundation for working with real-world datasets, where data is rarely perfectly organized from the beginning.

8.4 SQL Skills

SQL learning provides an understanding of how structured data can be retrieved and summarized from databases. This is particularly relevant for a CSE student because database systems are a fundamental part of many software applications. The ability to query information also supports later learning in business intelligence and data engineering.

8.5 Excel Skills

Excel provided a practical environment for organizing and examining data. Spreadsheet-based work makes it possible to quickly inspect records, perform calculations, filter information and create basic visualizations. These skills are useful for small analytical tasks and for preliminary inspection of datasets.

8.6 Data Visualization Skills

The introduction to Tableau reinforced the importance of communicating results visually. A technically correct analysis may still be ineffective if the audience cannot understand it. Visualization therefore becomes part of the communication process rather than simply a decorative final step.

8.7 Practical Understanding

The course connected concepts that are often learned independently: data characteristics, preparation, transformation, statistics, database querying and visualization. Seeing these as parts of one workflow made the purpose of each skill clearer.

8.8 Overall Learning Outcome

Table 8.1 – Major Learning Outcomes

Area	Learning Outcome
Data analytics process	Understand the stages through which raw data can become useful information
Data characteristics	Recognize differences in data types, formats and availability
Preparation	Understand common quality issues and preparation activities

Transformation	Understand how data can be reshaped for analysis
Statistics	Use basic statistical ideas to summarize information
Excel	Perform introductory spreadsheet-based analysis
SQL	Retrieve and summarize structured data
Tableau	Create and interpret visual representations of data
Communication	Present analytical work in a structured portfolio/report

CHAPTER 9 – SCOPE AND FUTURE APPLICATIONS

9.1 Scope of Data Analytics

The concepts learned in the course can be applied wherever structured information needs to be understood. Examples include business reporting, education, customer analysis, operations, finance, software products and performance monitoring. The specific analytical techniques will depend on the problem and the characteristics of the data.

9.2 Industry Applications

In business environments, analytics can support sales and customer analysis. In education, it can help summarize student performance and survey responses. In software products, usage data can be examined to understand patterns in application behaviour. In operational environments, data can support monitoring and comparison.

These examples demonstrate the broad relevance of the basic workflow, although the actual implementation would require domain-specific data, privacy considerations and analytical methods.

9.3 Future Learning

The course provides a foundation for further study rather than covering every area of data analytics. Possible next steps include more advanced SQL, Python-based data analysis, statistics, database design, data visualization, machine learning, data engineering and cloud-based analytics. Building these skills progressively can help connect introductory analytics with more advanced computing applications.

9.4 Career Relevance

Data-oriented skills can complement a computer science degree because many software systems generate and consume data. Knowledge of SQL, data preparation and visualization can be useful in roles involving software development, business intelligence, data analysis, database systems and product-oriented technology work. The course therefore adds a practical data perspective to a broader CSE skill set.

9.5 Possible Improvements

Future work can be improved by applying the learned workflow to larger and more complex datasets. Additional practice with real-world data would provide more experience with missing information, inconsistent records and domain-specific constraints. Further learning could also introduce automated data pipelines, programming-based analytics and more advanced statistical methods.

CHAPTER 10 – CONCLUSION

10.1 Conclusion

The Data Analytics Essentials course provided a useful foundation for understanding how data can be converted into meaningful information. The main learning was not limited to individual software tools. Instead, the course presented data analytics as a process involving understanding the data, acquiring relevant information, preparing it carefully, transforming it where necessary, analyzing it, visualizing important findings and communicating the results.

The training also helped establish the relationship between different technical tools. Excel provides a practical environment for organizing and examining data, SQL provides a structured way to retrieve and summarize database information, and Tableau supports visual communication of analytical findings. These tools serve different purposes, but they can contribute to the same overall workflow.

Another important learning outcome was the importance of data quality. Analysis performed on incomplete, duplicated or inconsistent information can produce misleading results. For this reason, preparation and validation should be treated as essential parts of analytics rather than optional preliminary steps. The course also encouraged a more careful approach to interpretation. A pattern in a dataset needs to be considered in context, and a visualization should communicate evidence without creating conclusions that the data cannot support.

From the perspective of a B.Tech CSE student, the training is relevant because data is closely connected to software systems, databases and applications. A computer science professional may work with data at several stages, from storing it in databases to retrieving, processing and presenting it. Understanding analytics provides an additional layer of knowledge that can help connect technical implementation with the information that users and organizations actually need.

The course also provided a starting point for further technical development. The concepts of data preparation, SQL and visualization can be extended through more advanced database systems, Python-based analytics, statistics, machine learning, business intelligence and cloud-based data platforms. Continued practice with real datasets would strengthen the ability to handle the complexities that occur in practical environments.

Overall, the Data Analytics Essentials training strengthened my understanding of the data analytics lifecycle and gave me practical exposure to important analytics tools. It helped me understand that useful analysis is not simply about producing a graph or calculating an average. It begins with asking the right question, working with appropriate data, preparing that data carefully, selecting suitable methods and finally communicating the result in a clear and responsible way. This understanding will be useful in my further academic work and in future technical projects involving data and software systems.

10.2 Overall Experience

The training experience was valuable because it brought together conceptual and practical aspects of data analytics. Working through the different areas made it easier to understand how individual activities fit into a larger process. The exposure to multiple tools also demonstrated that analytics is not dependent on a single software application; the choice of tool depends on the nature of the task and the data.

10.3 Final Learning Summary

The most significant outcome of the training is a stronger foundation in data-oriented thinking. I developed an understanding of the analytics process, data characteristics, acquisition, preparation, transformation, basic statistics, Excel, SQL, Tableau and portfolio communication. These skills form a useful base for future learning and practical work in computer science and data-related areas.

CHAPTER 11 – CERTIFICATES AND SUPPORTING DOCUMENTS

The Cisco Networking Academy Certificate of Course Completion for Data Analytics Essentials is attached as supporting evidence of the training completed for this report. The certificate is issued in the name of Sreedevi EJ and records successful completion of the Data Analytics Essentials course.

The certificate identifies the areas of proficiency associated with the course, including the data analytics process, data characteristics, data transformation, basic statistical and data preparation techniques, hands-on work using Excel, SQL and Tableau, and project portfolio evaluation.

The original certificate should be appended to the final printed report according to the submission requirements of the university.



CHAPTER 12 – BIBLIOGRAPHY / REFERENCES

11. Cisco Networking Academy. Data Analytics Essentials. Certificate of Course Completion issued to Sreedevi EJ, 11 June 2026.
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13. Microsoft Excel. Spreadsheet and data analysis software documentation and learning resources.
14. SQL. Structured Query Language concepts for relational data querying and analysis.
15. Tableau. Data visualization and analytics learning resources.